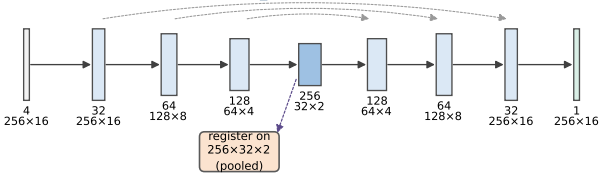
FNO, TRL2D arena (The Well; 19.0 M parameters) -- spectral operator



CNextU-net on rayleigh_benard, shear_flow, MHD_64 (released; architecture as Fig. 1)

rayleigh_benard: 16×512×128 → stem 42 → E1-E4 84/168/336/672 → neck 672×32×8 (register, pooled) → D4-D1 → 4×512×128 (18.6 M)
shear_flow: 16×256×512 → neck 672×16×32 → 4×256×512 (18.6 M). MHD_64, 3-D convs: 28×64^3 → neck 672×4^3 → 7×64^3 (22.3 M)

U-Net, 1HP heat plume (1HP_NN; 7.8 M parameters) -- convolutional encoder-decoder



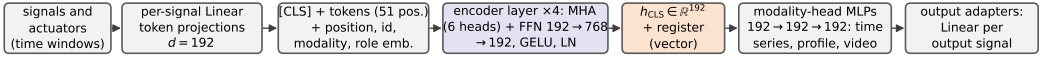
encoder: [conv 5×5, BN, ReLU]×2 per level, max-pool 2; decoder: convT 2×2/2, concat skip (dashed), conv blocks; head: conv 1×1 → temperature 1×256×16

UNetLite, Sophelio starter (3.1 M parameters) -- MLP into a transposed-convolution decoder

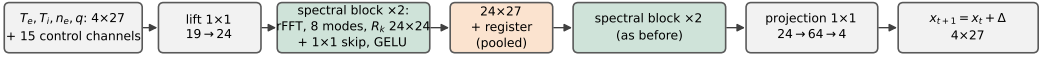


convT 4×4/2 + BN + ReLU (three), conv 2×2 → ψ; skip: fc 21 → 32,768 reshaped to 128×16×16, added after the first convT

TokaMind, TokaMark (tokamind-base-v2) -- transformer encoder



1-D spectral network, TORAX-Bench (27 radial points) -- 1-D Fourier operator



Residual MLP trunk, Fusion Equilibrium (17.8 M parameters; app. "Scores relative to published numbers")

